Attribute Types

- **Nominal:** categories, states, or "names of things"
  - Hair color = {auburn, black, blond, brown, grey, red, white}
  - marital status, occupation, ID numbers, zip codes
- **Binary**
  - An important Nominal attribute with only 2 states (0 and 1)
  - Symmetric binary: Both outcomes equally important
    - e.g., gender
  - Asymmetric binary: outcomes not equally important
    - e.g., medical test (positive vs. negative)
    - Convention: assign 1 to most important outcome (COVID positive)
- **Ordinal**
  - Values have a meaningful order (ranking) but magnitude between successive values is not known
  - Size = {small, medium, large}, grades, army rankings

Numeric Attribute Types

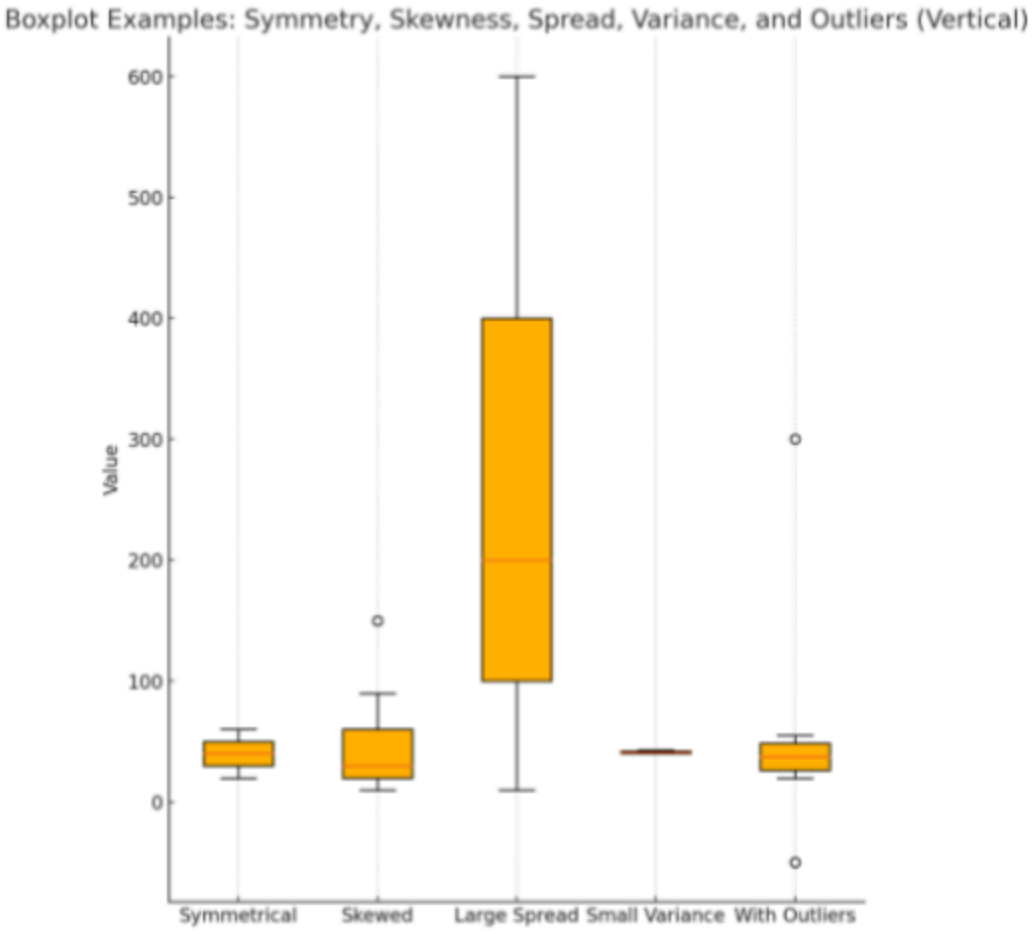
- **Interval**: Numeric
  - Measured on a scale of equal-sized unites
  - Values have order
    - E.g., temperature in C or F, calender dates
  - **No true zero-point**
- **Ratio** Numeric
  - Inherent **zero point**
  - We can speak of values as being an order of magnitude larger than unit of measurement (10K is twice as high as 5K)a
    - e.g., temperature in Kelvin, length, counts, monetary quantities

Properties of Attribute Values

- Distinctness: = !=
- Order: <>
- Addition: -+
- Multiplication: */
- Nominal: Distinctness
- Ordinal: Distinctness and order
- Interval: Distinctness, order, and addition
- Ratio: All 4

Boxplot Analysis

- Sort data in ascending order
- Calculate Median (Could be average of two middle #)
- Calculate Q1 and Q3 (Take median of the lower/higher half)
- Calculate IQR = Q3 - Q1
- Lower bound = (Q1 - 1.5 IQR)
- Upper bound = (Q3 + 1.5 IQR)



Data Matrix and Dissimilarity Matrix

- Data Matrix (object-by-attribute structure)
  - n data points with p dimensions

- two modes

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

- Dissimilarity matrix (object-by-object structure)
  - n data points, but registers only the distance
  - A triangular matrix
  - One mode

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

- d(i,j) is the measured dissimilarity between objects i and j
- Is close to 0 when i and j are highly similar or near each other
- sim(i,j) = 1 - d(i,j) (not always)

Minkowski Distance

- Assume our data objects have p numeric attributes

$$d(i,j) = (\sum_{k=1}^{p} |x_{ik} - x_{jk}|^r)^{\frac{1}{r}} \; x_{ik}\, x_{jk} \text{ are the kth element of a data object } x_i \text{ and } x_j, r >= 1$$

- r = 1: **Manhattan distance** (city block, L1 norm)
  - Hamming distance: the number of bits that are different between binary vectors
- r = 2: **Euclidean distance** (L2 norm)
- r = ∞: **Supremum distance** (Lmax norm, L_∞ norm)

- d(i,j) = $max_{1<=k<=p}|x_{ik} - x_{jk}|$

Example:

x = (1,2)

y = (4,6)

1. Manhattan Distance (r = 1)
   $|1-4| + |2-6| = 3 + 4 = 7$
2. Euclidean Distance (r = 2)
   $\sqrt{[(1-4)^2 + (2-6)^2]}$
   $= \sqrt{(9 + 16)}$
   $= 5$
3. Supremum Distance (r = ∞)
   $max(|1-4|, |2-6|)$
   $= max(3,4)$
   $= 4$

Similarity Between Binary Attributes:

- Similarity measures between objects that contain only binary attributes are called similarity coefficients and typically have values between 0 and 1
- **Simple Matching Coefficient (SMC)** counts both presence and absences equally

$$sim(i,j) = \frac{\#\text{ of matching attributes}}{\#\text{ of attributes}} = \frac{f_{11} + f_{00}}{f_{00} + f_{01} + f_{10} + f_{11}}$$

- **Jaccard Coefficient (J)** counts presences and it is frequently for asymmetric binary attributes

$$sim(i,j) = \frac{\#\text{ of 11 matches}}{\#\text{ of not - both - zero attributes}} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

- Cosine measure: If d_1 and d_2 are two vectors

$$cos(d_1, d_2) = \frac{d_1 \cdot d_2}{||d_1|||d_2|}, \text{ dot product on top}$$

Example:

d1 = (1, 2, 3)

d2 = (2, 4, 6)

Dot product:

(1)(2) + (2)(4) + (3)(6) = 28

Magnitudes:

$\|d1\| = \sqrt{(1^2+2^2+3^2)} = \sqrt{14}$

$\|d2\| = \sqrt{(2^2+4^2+6^2)} = \sqrt{56}$

Cosine similarity:

$28 / (\sqrt{14} \times \sqrt{56}) = 28 / 28 = 1$

Interpretation: Perfect positive similarity (same direction).

Noisy Data

- Binning
  - First sort data and partition into (equal- frequency) bins
  - Then one can smooth by bin means, smooth by bin median smooth by bin boundaries, etc.
    - Equal-width (distance) partitionin
      - Divides the range into *N* intervals of equal size
      - If *A* and *B* are the lowest and highest values of the attribute, the width of the intervals will be $W = \frac{B-A}{N}$
      - Skewed data not handled well
    - Equal-depth (frequency) partitioning
      - Divides the range into *N* intervals, each containing approx same number of samples
      - Good data scaling
      - Managing categorical attributes can be tricky
- Regressions
  - smooth by fitting the data into regression functions
- Clustering
  - detect and remove outliers
- Combined computer and human inspection
  - detect suspicious values and check by human

Correlation

- X^2 (chi- squared) test

$$X_{ij} = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

$$e_{ij} = \frac{count(A = a_i) * count(B = b_i)}{n}$$

Q: Are gender and preferred reading correlated?

	male	female	Sum (row)
Like science fiction	250(90)	200(360)	450
Not like science fiction	50(210)	1000(840)	1050
Sum(col.)	300	1200	1500

- X^2 (chi-square) calculation (numbers in parenthesis are expected counts calculated based on the data distribution in the two categories)
  - E.g. 90 = (300 x 450) / 1500

$$\chi^2 = \frac{(250-90)^2}{90} + \frac{(50-210)^2}{210} + \frac{(200-360)^2}{360} + \frac{(1000-840)^2}{840} = 507.93$$

- Correlation coefficient:

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{n\sigma_A\sigma_B}$$

Where n is # of tuples, $\bar{A}$ and $\bar{B}$ are respective means of A and B, σ_A σ_B are respective stand deviation

- Note that -1 <= $r_{A,B}$ <= 1
- If $r_{A,B}$ > 0, A and B are positively correlation. The higher, the stronger
- $r_{A,B}$ = 0: independent
- $r_{A,B}$ < 0: negatively correlated

Step 2 : Create a Working Table

A	B	$A - \bar{A}$	$B - \bar{B}$	Product
1	2	-2	-2	4
2	4	-1	0	0
3	5	0	1	0
4	4	1	0	0
5	5	2	1	2

$$\sum (A - \bar{A})(B - \bar{B}) = 4 + 0 + 0 + 0 + 2 = 6$$

Step 3 : Compute Standard Deviations

$$\sigma_A = \sqrt{\frac{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}{5}} = \sqrt{\frac{10}{5}} = \sqrt{2}$$

$$\sigma_B = \sqrt{\frac{(-2)^2 + 0^2 + 1^2 + 0^2 + 1^2}{5}} = \sqrt{\frac{6}{5}}$$

Step 4 : Correlation Formula

$$r = \frac{\sum (A - \bar{A})(B - \bar{B})}{n \sigma_A \sigma_B}$$

$$r = \frac{6}{5 \cdot \sqrt{2} \cdot \sqrt{6/5}}$$

$$r \approx 0.77$$

Covariance

- It can be simplified in computation as

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n} \qquad Cov(A,B) = E(A\cdot B) - \bar{A}\bar{B}$$

- Suppose two stocks A and B have the following values in one week:

$$(2, 5), (3, 8), (5, 10), (4, 11), (6, 14).$$

- Question: If the stocks are affected by the same industry trends, will their prices rise or fall together?
  - $E(A) = (2 + 3 + 5 + 4 + 6)/\ 5 = 20/5 = 4$
  - $E(B) = (5 + 8 + 10 + 11 + 14)\ /5 = 48/5 = 9.6$
  - $Cov(A,B) = (2\times5+3\times8+5\times10+4\times11+6\times14)/5 - 4 \times 9.6 = 4$
- Thus, A and B rise together since $Cov(A, B) > 0$.

- Diagonal elements are variances Cov(x,x) = var(x)
- Covariance Matrix is symmetric
- It is positive semi-definite matrix

Normalization

- **Min-max normalization**: to [new_min$_A$, new_max$_A$]

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

  - Ex. Let income range \$12,000 to \$98,000 normalized to [0.0, 1.0]. Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000}(1.0 - 0) + 0 = 0.716$

- **Z-score normalization** (μ: mean, σ: standard deviation):

$$v' = \frac{v - \mu_A}{\sigma_A}$$

  - Ex. Let μ = 54,000, σ = 16,000. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- **Normalization by decimal scaling**

$$v' = \frac{v}{10^j}$$ Where j is the smallest integer such that Max(|v'|) < 1

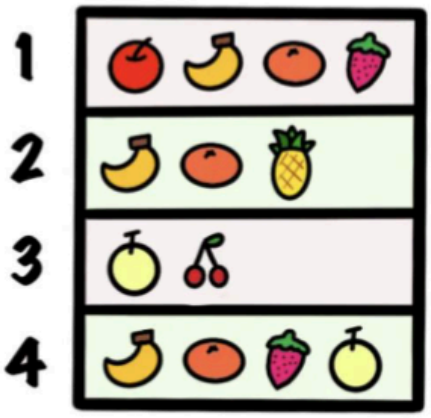
Frequent Patterns

- Find combinations of items (itemsets) that occure frequently
  - Called Frequent itemset or frequent patterns
- Support Count: Frequency or occurrence of an itemset X
- Relative Support: fraction of transactions that contains X (probability that a transaction contains X)

- An itemset X is frequent if X's support is no less than a *minsup* threshold

Item set: (apple, banana)

Dataset:



Support (relative support) is the **proportion of transactions containing a certain item set**. An item set can contain more than one item.

Support (apple, banana) = 1/4

Confidence is the **proportion of transactions containing A also contains B**.

$$\text{Confidence of } A \rightarrow B = \frac{\text{Support}(A \cup B)}{\text{Support}(A)}$$

Confidence of (apple -> banana) = 1

Level-wise approach

C_k = candidate itemsets of size k
L_k = frequent itemsets of size k

1. k = 1, C_1 = all items
2. While C_k not empty

Frequent itemset generation

3. Scan the database to find which itemsets in C_k are frequent and put them into L_k

Candidate generation

4. Use L_k to generate a collection of candidate itemsets C_{k+1} of size k+1
5. k = k+1

Database TDB

Tid	Items
T100	I1,I2,I5
T200	I2,I4
T300	I2,I4
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I2,I3

$Sup_{min} = 2$

C_1 — 1st scan →

Itemset	sup
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

L_1 — compare →

Itemset	sup
{I1}	6
{I2}	7
{I3}	6
{I4}	2
{I5}	2

C_2

Itemset
{I1, I2}
{I1, I3}
{I1, I4}
{I1, I5}
{I2, I3}
{I2, I4}
{I2, I5}
{I3, I4}
{I3, I5}
{I4, I5}

2nd scan →

Itemset	sup
{I1, I2}	4
{I1, I3}	4
{I1, I4}	1
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2
{I3, I4}	0
{I3, I5}	1
{I4, I5}	0

C_2 — compare →

Itemset	sup
{I1, I2}	4
{I1, I3}	4
{I1, I5}	2
{I2, I3}	4
{I2, I4}	2
{I2, I5}	2

L_2

C_3

Itemset
{I1, I2, I3}
{I1, I2, I5}

3rd scan →

L_3

Itemset	sup
{I1, I2, I3}	2
{I1, I2, I5}	2

- We have all frequent itemsets, how do we get the rules?
  - For every frequent itemset S, we consider all subsets $L \subset S$, and we find rules of the form $S - L \rightarrow L$, that satisfy the minimum confidence requirement
  - Example: $S = \{A,B,C,D\}$
  - Candidate rules:

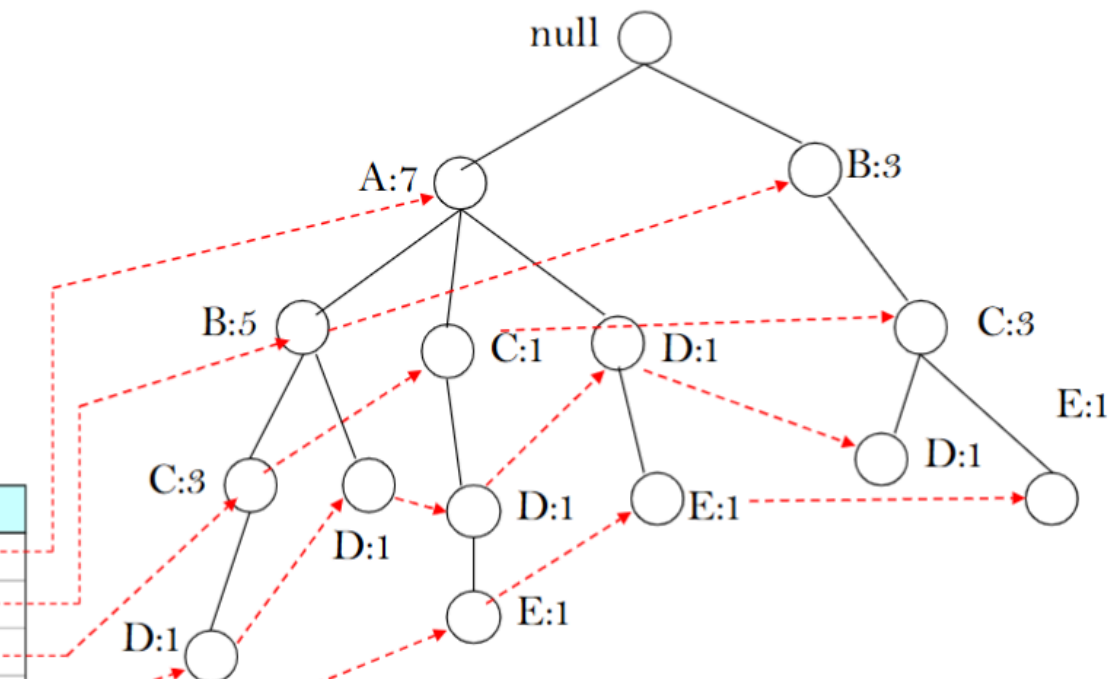
$BCD \rightarrow A$, $ACD \rightarrow B$, $ABD \rightarrow C$, $ABC \rightarrow D$,
$CD \rightarrow AB$, $BD \rightarrow AC$, $BC \rightarrow AD$, $AD \rightarrow BC$, $AB \rightarrow CD$, $AC \rightarrow BD$, $D \rightarrow ABC$, $C \rightarrow ABD$, $B \rightarrow ACD$, $A \rightarrow BCD$

- If $|S| = k$, then there are $2^k - 2$ candidate association rules (ignoring $S \rightarrow \varnothing$ and $\varnothing \rightarrow S$)

TID	Items
1	{A,B}
2	{B,C,D}
3	{A,C,D,E}
4	{A,D,E}
5	{A,B,C}
6	{A,B,C,D}
7	{B,C}
8	{A,B,C}
9	{A,B,D}
10	{B,C,E}

Transaction Database

Each transaction is a path in the tree



Header table

Item	Pointer
A	
B	
C	
D	
E	

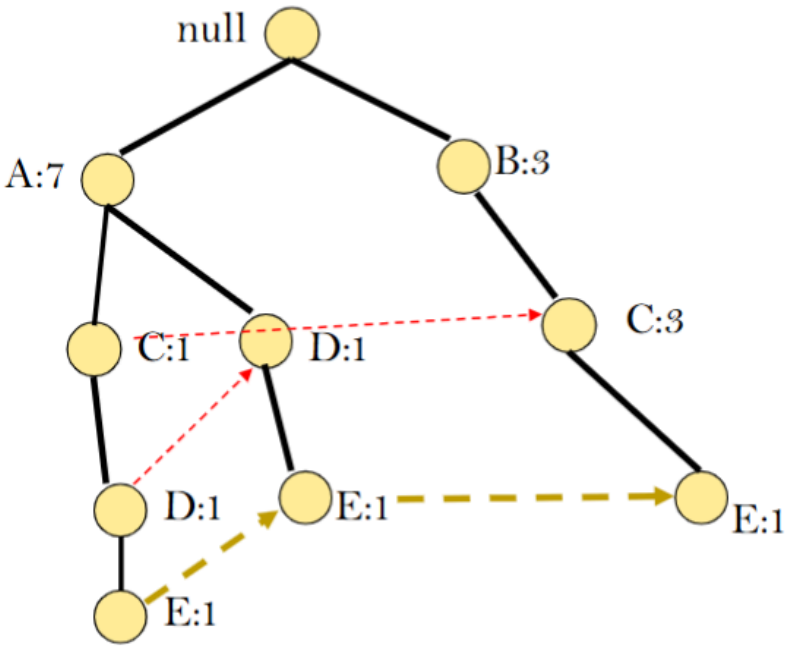
Pointers are used to assist frequent itemset generation

Recompute Support

The support counts for some of the nodes include transactions that do not end in E

For example in null->B->C->E, we count {B,C}

The support of any node is equal to the sum of the support of leaves with label E in its subtree



$Sup_{min} = 2$

Database TDB

Tid	Items
10	A, C, D
20	B, C, E
30	A, B, C, E
40	B, E

1st scan

itemset	TID Set
A	{T10,T30}
B	{T20,T30,T40}
C	{T10,T20,T30}
D	{T10}
E	{T20,T30,T40}

Prune

itemset	TID Set
A	{T10,T30}
B	{T20,T30,T40}
C	{T10,T20,T30}
E	{T20,T30,T40}

Intersecting

itemset	TID Set
AB	{T30}
AC	{T10,T30}
AE	{T30}
BC	{T20,T30}
BE	{T20,T30,T40}
CE	{T20,T30}

Prune

itemset	TID Set
AC	{T10,T30}
BC	{T20,T30}
BE	{T20,T30,T40}
CE	{T20,T30}

Intersecting

itemset	TID Set
{B, C, E}	{T20,T30}

(A given 3-itemset is a candidate 3-itemset only if every one of its 2-itemset is frequent)